Lab 05 – Reflection Journal
ITAI 1371 – Introduction to Machine Learning
Group 4

Lab Topic: Data Preparation

Member 1: Patrick Rostand Gandjouon Tchassem.

In this lab, I learned how important data preparation is before building a machine learning model. Working with the Titanic dataset helped me understand that real-world data is not always complete or ready to use. One thing I found interesting was learning how to handle missing values using the median instead of deleting rows. Our dataset had 177 missing values in the Age column, and after applying the code, the output showed zero missing values.

I also learned how One-Hot Encoding converts categorical information, such as Sex and Embarked, into numerical features that machine learning models can understand. Another important part was using StandardScaler to bring Age and Fare values onto a similar scale. At first, some of these concepts were confusing, but reviewing the code and outputs helped me understand them better.

Overall, this lab helped me see why data preparation is an important step in machine learning. I now understand that cleaning and preparing data properly can help improve the quality of a machine learning model.

Member 2: Saimi Manasiya

In this lab, I learned about the importance of data preparation in machine learning using the Titanic dataset. I understood how to handle missing values, encode categorical data, and scale numerical features. I learned that preparing data properly is an important step before building a machine learning model because the quality of the data can affect the results. This lab also helped me understand how Python and pandas can be used to clean and organize a dataset.

I also gained a better understanding of how different data preparation techniques work and why they are needed. I learned that missing values can be handled using the median, categorical data can be converted into numerical form through encoding, and numerical features can be scaled using StandardScaler. Overall, this lab improved my practical knowledge and helped me feel more confident working with datasets. It also taught me the importance of teamwork, communication, and learning from other group members.

Member 3: Kenneth Kouokam

For Module 05, my assigned role is Dataset Setup and Initial Review. The main idea I take from this part of the lab is that loading a dataset is only the beginning of preparing it. A table can look organized while still containing missing information or columns that need to be transformed. Before thinking about a model, I need to understand what the data contains and what might make it difficult to use. This gives the setup stage more purpose than simply getting the notebook started.

Missing values are an important example. A blank age does not mean that a passenger was zero years old; it means that the information is unavailable. That difference matters because replacing a missing value with an arbitrary number could change the meaning of the data. Checking missing-value counts first helps explain why a later step fills missing ages with the median. It also reminds me that filling a gap produces an estimate, not the passenger's actual age. A complete-looking table is not automatically an accurate one.

Column types are another part of the initial review that connects directly to the rest of the lab. Age
and Fare represent numerical measurements, while Sex and Embarked describe categories. These

columns need different treatment, so I cannot apply the same preparation step to everything. Encoding gives categories a form that many models can use, while scaling changes the numerical range of selected measurements. Understanding these differences helps me see the notebook as a sequence of decisions rather than a list of commands to copy.

This also connects to the exploratory data analysis from the previous module. Exploring data helps reveal patterns, but it can also reveal problems that need attention before further analysis. For my role, the useful question is not just whether the file loads, but whether the group understands the starting data. Recording the original missing-value counts will give us something to compare with the later results. That comparison is a simple way to check whether a preparation step actually did what we intended.

As I carry out my section, I want to focus on explaining what each check tells me instead of only looking for an error-free output. One question I want to keep in mind is whether different columns need different approaches to missing information. I also want my notes to be clear enough for the next group member to use. My takeaway is that careful preparation begins with careful observation: understanding the starting data makes the later choices easier to explain and check.

Member 4: Anavictoria Cavazos

Reflection for Module 05

For Module 05 I learned the different steps and parts in preparing data, such as the importance of the median,
one-hot encoding, and the difference in certain features. Starting with the importance of the median in data when there
are missing values. When it comes to missing values/numbers in data you still want the best accuracy of your data, so
when you use the median instead of the mean, your outcomes have a higher chance of being accurate because the outliers
are not affecting the outcome since they are being overlooked in this part.

When I did the one-hot encoding I had to relook at the steps that I had just done to really notice the process to understand
fully what one-hot encoding is. Based on the one-hot encoding that I practiced in the assignment, one-hot encoding is taking in a
text or words where this type of coding adjusts to 0 and 1, for true or false, so that the data can be processed as numbers and have
an output instead of an error. This helped me see and understand the language for certain parts of the data and how some coding
sometimes has to be specific and fixed in order for it to be able to be imputed with a good output.

In addition, in some of the steps that I used through the data preparation, I noticed how each of the features functioned
in ways that worked according to the data and expected outcomes. One of these steps was feature scaling, which
is applied when features may range in numbers that can add importance to a feature that is not actually if importance
simply because it has a larger number. This was seen in step/ task 3, where the method standardization was used for 'age'
and 'fare' because both are seen in forms of number. Aside from completing this task, this part helped me understand that
sometimes in data, you have to shorten it using certain codes, so then the data is shown in a proper way, which is done
during proper preparation for a proper/expected output.

Member 5: Hashim Sayed Hoosini

Course: 1371- Intro to Machine Learning

Professor: Viswanatha Rao

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Data preparation is a critical and time-consuming stage of every machine learning project. This lab perfectly demonstrated why real-world data is rarely model-ready and showed how we systematically prepare data for machine learning pipelines. Through this lab, I dove deep into the data preparation process, which gave me a better understanding of each technique and concept. I worked with the Titanic dataset, which I previously had the chance to work with in another project. It contains passenger information but is riddled with missing values. The goal of this project was to go through the data preprocessing steps and clean the data thoroughly for machine learning projects.

Phase 1: Initial Data Ingestion and Exploratory Visualization

First, I imported the necessary libraries, such as pandas and numpy, and loaded the Titanic dataset directly from a public URL using pd.read_csv(). Then, I displayed the first five rows of the dataset using display(df.head()). I noticed that there were several missing values in the dataset. In order to better analyze the data before cleaning it, I used matplotlib and seaborn to visualize the data through a missing value heatmap and a count of missing values per column.

Phase 2: Missing Data Mitigation and Strategic Imputation

Second, I had to handle the missing values and drop redundant columns, because most machine learning models cannot handle missing data. I dropped the redundant text columns that were missing too many values. I also used imputation techniques, which means filling missing values with a calculated guess, to fill the remaining empty columns. Specifically, I used the median age to fill the missing Age values and the mode to fill the missing Embarked port values. Afterward, I verified the Age column imputation to ensure that there were no missing values remaining.

Phase 3: Categorical Feature Transformation via One-Hot Encoding

Then, I had to encode the categorical features. Because machine learning models are mathematical, they need numbers instead of text. I converted categorical columns such as Sex and Embarked into a numerical format using the one-hot encoding method. This method turns categorical values into numerical values like 1 and 0, and transforms the text fields into binary True and False indicators. Finally, I verified the encoded features to ensure Sex and Embarked columns successfully transformed into clean numerical indicators.

Phase 4: Feature scaling concept:

Models are sensitive to the scale of features. For instance, age (0-80) and fare (0-512) are on two different scales. This can cause the model to be biased toward fare; because fare has bigger numbers, the model will assume fare is more important. To solve this issue, we use feature scaling. Feature scaling puts all the values on the same scale. Standardization is a common method which uses StandardScaler in Scikit-learn. This puts all the numbers to have a mean of zero and a standard deviation of 1.

Phase 5: Post-Preprocessing Diagnostic Analysis and Insights

For plotting post-preprocessing insights, I used code to visualize two updated diagnostic charts: the new missing value heatmap and a correlation matrix of the numerical and encoded features. This insight confirms that no missing values remain in the dataset. Additionally, the correlation heatmap measures the strength of the relationships between all numerical features.

Challenges: For this lab, I faced different challenges, from code issues and indentation errors to searching for ‘Age’ or ‘Sex’ instead of ‘age’ and ‘sex’, which I had to use the code copilot to overcome. These debugging experiences and working through these issues helped me to develop my diagnostic skills. Furthermore, to develop my skills beyond the assignment, I split the dataset into train and test sets. Splitting the dataset into 80% training data and 20% testing data allows us to evaluate the performance on unseen data.

Overall, this lab showed me the gap between raw and messy data and structured model deployment. Through this lab, I dove deep into the data preparation process, which provided me a better understanding of each technique. Now I feel confident using these techniques in future projects. I

will always follow each step to build a clean, modular preprocessing pipeline.

LLM used:

GoogleColab copilot and Google AI mode

Purpose: writing the code and fixing spelling and grammar errors.
